Digital Systems

Tutorial 11

March-June 2023

1. Investigation the transition table of the figure below and determine all race conditions and whether they are critical or non-critical.

$y_1 y_2$ \ $x_1 x_2$	00	01	11	10
00	00	11	10	01
01	10	01	10	11
11	11	11	10	10
10	11	01	10	10

Solution:

(a) In row y_1y_2=00, transition from $x_1x_2 = 00 \rightarrow x_1x_2$=01:
possible transitions
00 $\rightarrow$ 11
00 $\rightarrow$ 10 $\rightarrow$ 01
00 $\rightarrow$ 01
stable states are different $\Rightarrow$ critical race condition

(b)In row $y_1y_2 = 01$, transition from $x_1x_2 = 01 \rightarrow x_1x_2 = 00$:
possible transitions
01 $\rightarrow$ 10 $\rightarrow$ 11
01 $\rightarrow$ 00
01 $\rightarrow$ 11
stable states are different $\Rightarrow$ critical race condition

(c) In row $y_1y_2 = 01$, transition from $x_1x_2 = 01 \rightarrow x_1x_2 = 11$:
possible transitions
01 $\rightarrow$ 10
01 $\rightarrow$ 00 $\rightarrow$ 01
01 $\rightarrow$ 11 $\rightarrow$ 01
stable states are same $\Rightarrow$ non-critical race condition

(d) In row $y_1y_2 = 10$, transition from $x_1x_2 = 11 \rightarrow x_1x_2 = 01$:
possible transitions

10 → 01
10 → 00 → 11
10 → 11
stable states are different ⇒ critical race condition

In following table, all races and type of races are shown.

$y_1 y_2$ \ $x_1 x_2$	00	01	11	10	
00	00	11	10	01	Critical
01	10	01	10	11	Critical Non-critical race
11	11	11	10	10	
10	11	01	10	10	Critical

2. Determine all race conditions and are they critical or non-critical.

$y_1 y_2$ \ $x_1 x_2$	00	01	11	10
00	10	00	11	10
01	01	00	10	10
11	01	00	11	11
10	11	00	10	10

Solution:

(a) In row y_1y_2=00, transition from $x_1x_2 = 01 \rightarrow x_1x_2$=11:
possible transitions
00 → 11
00 → 10
00 → 01 → 10
stable states are different ⇒ critical race condition

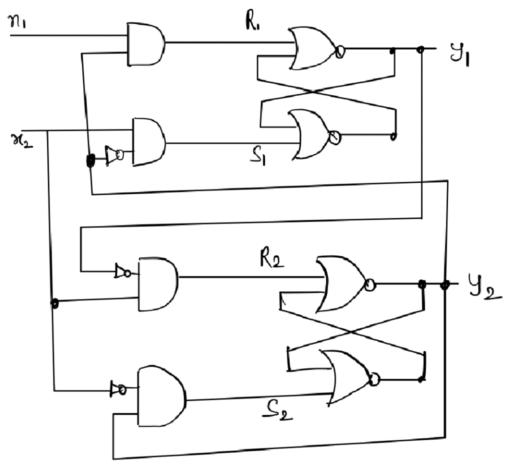
(b) In row y_1y_2=11, transition from $x_1x_2 = 11 \rightarrow x_1x_2$=01:
possible transitions
11 → 00
11 → 10 → 00
11 → 01 → 00
stable states are same ⇒ non critical race condition

(c) In row y_1y_2=01, transition from $x_1x_2 = 00 \rightarrow x_1x_2$=10:
possible transitions
01 → 10
01 → 00 → 10
01 → 11 → 11
stable states are different ⇒ critical race condition

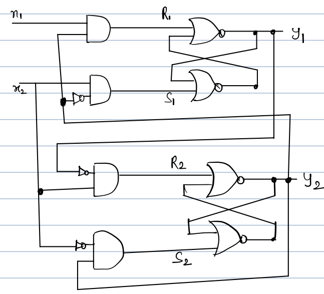
In the following table, all races and types of races are shown.

$y_1 y_2$ \ $x_1 x_2$	00	01	11	10	
00	10	00	11	10	Critical race
01	01	00	10	10	Critical race
11	01	00	11	11	Non critical race
10	11	00	10	10	

3. Analyze the SR latch circuit and
   (a) derive the transition table (b) race condition



Solution:



The boolean function of S and R inputs,

$S_1 = x_2 y_2'$ $\qquad$ $S_2 = x_2' y_2$

$R_1 = x_1 y_2$ $\qquad$ $R_2 = x_2 y_1'$

We check the $SR = 0$, to ensure the proper operation of circuit,

$S_1 R_1 = x_2 y_2' \cdot x_1 y_2 = 0$

$S_2 R_2 = x_2' y_2 \cdot x_2 y_1' = 0$

The excitation function are derived from relation,

$Y = S + R'y$

$Y_1 = x_2 y_2' + (x_1 y_2)' y_1$

$\quad = x_2 y_2' + x_1' y_1 + y_2' y_1$

$Y_2 = x_2' y_2 + y_1 y_2$

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	1	1	0	0
10	1	1	1	1

Map for Y_1

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	1	1	1
10	0	0	0	0

Map for Y_2

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	00	10	10	00
01	01	00	00	01
11	11	11	01	01
10	10	10	10	10

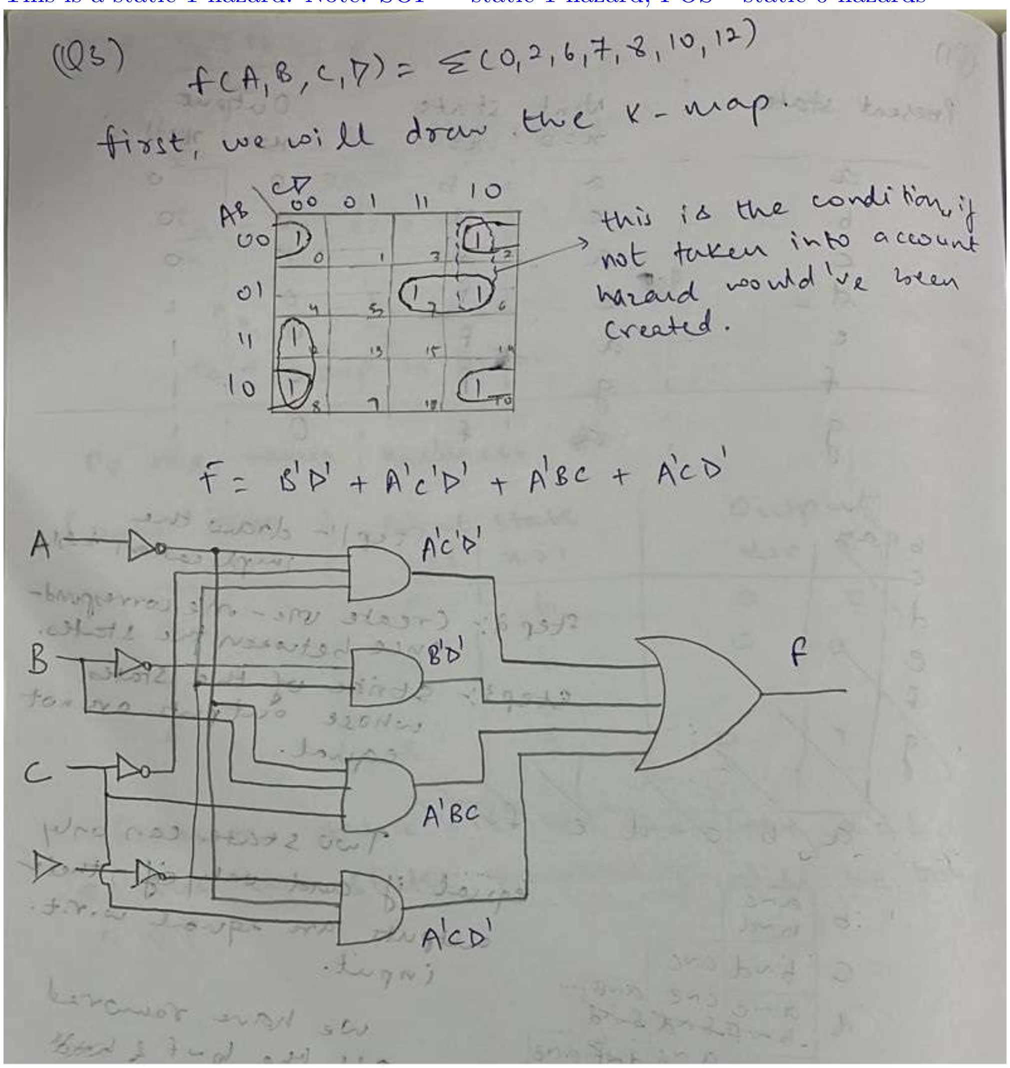
Transition table

- There is no race

4. Find a circuit that has no static hazards and implements the boolean functions.
$F(A, B, C, D) = \Sigma(0, 2, 6, 7, 8, 10, 12)$.

Solution:

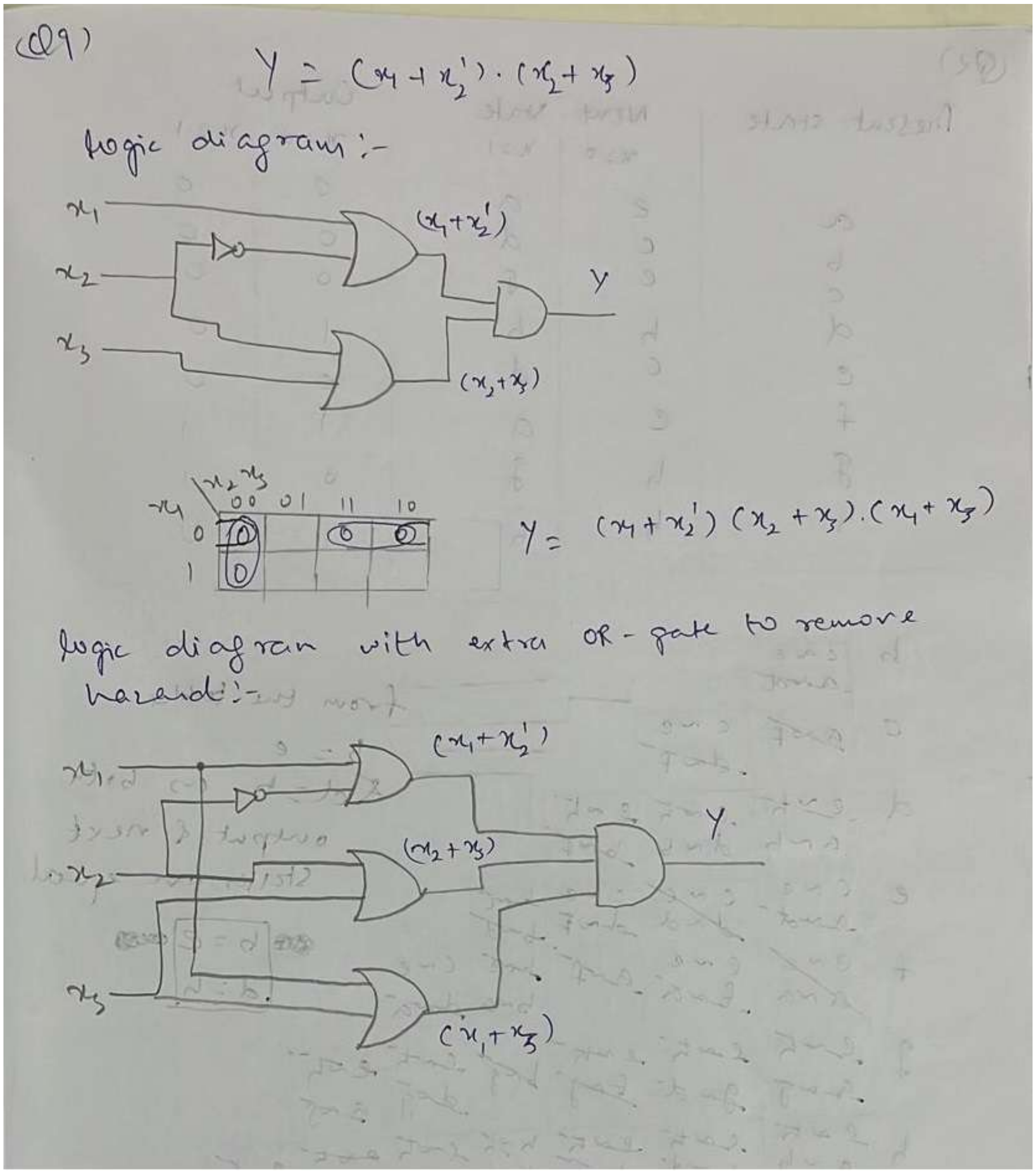
This is a static-1 hazard. Note: SOP = static-1 hazard, POS= static-0 hazards



5. Draw the logic diagram of the product of sums expression, $Y = (x_1 + x_2').(x_2 + x_3)$,find a way

to remove the hazard by adding one more OR gate.

Solution:



6. Obtain a primitive flow table for a circuit with two inputs, x_l and x_2, and one output, z, that satisfy the following two conditions:

(a) The inputs x_l and x_2 never changes simultaneously.

(b) The output z is complemented if $x_l = 1$ and the x_2 changes from 0 to 1. Otherwise, the output z remains unchanged under any other input condition.

Solution:

Given,

x_1 & $x_2 \rightarrow$ inputs $\quad z \rightarrow$ output

$x_1 = 1$ & $x_2 = 0 \rightarrow 1 \Rightarrow z = \bar{z}$ { Output is complimental

other i/p cond-n $\Rightarrow z = z$ { No change

x_1 & x_2 = Not change simultaneously

- Specification of total state :-

State	Input x_1	Input x_2	Output z	Comments	Next state (NS)
a	1	0	0	Initial output $z=0$	NS: e or b
b	1	1	1	After state a	NS: h or c
c	1	0	1	Initial output $z=1$	NS: g or d
d	1	1	0	After state c	NS: f or a
e	0	0	0	After states a or f	NS: f or a
f	0	1	0	After state d or e	NS: e or d
g	0	0	1	After states c	NS: h or c
h	0	1	1	After states b or g	NS: g or b

• Primitive flow table:

	$x_1 x_2$			
	00	01	11	10
a	e, –	–, –	b, –	a, 0
b	–, –	h, –	b, 1	c, –
c	g, –	–, –	d, –	c, 1
d	–, –	f, –	d, 0	a, –
e	e, 0	f, –	–, –	a, –
f	e, –	f, 0	d, –	–, –
g	g, 1	h, –	–, –	c, –
h	g, –	h, 1	b, –	–, –

7. Obtain a primitive flow table for a fundamental mode operated circuit meeting the following requirements:

   (a) There are two inputs, x_l and x_2, and single output, z.

   (b) The output z is always 0, when x_l= 0, independent of the x_2 value.

   (c) The output z becomes 1 if x_2 changes while x_1=1 and it remains until x_1 becomes 0 again.

Solution:

Given,

x_1 & $x_2 \longrightarrow$ inputs $\qquad$ $z \rightarrow$ output

$x_1 = 0 \Rightarrow z = 0$

$x_1 = 1$ & $x_2 = \begin{matrix} 0 \rightarrow 1 \\ \text{or} \\ 1 \rightarrow 0 \end{matrix} \Rightarrow z = 1$

- Specification of total states:

State	Input x_1	Input x_2	Output z	Comments	Next state (NS)
a	1	0	0	initial output z=0	NS: d or b
b	1	1	1	After state a	NS: d or c
c	1	0	1	After state b ~~or a~~	NS: f or b
d	0	1	0	After state b or a	NS: f or e
e	1	1	0	After state d	NS: d or c
f	0	0	0	After states d, e or c	NS: d or a

- Primitive flow table:

x_1x_2

	00	01	11	10
a	-,-	d,-	b,-	a,0
b	-,-	d,-	b,1	c,-
c	f,-	-,-	b,-	c,1
d	f,-	d,0	e,-	-,-
e	-,-	d,-	e,0	c,-
f	f,0	d,-	-,-	a,-

8. Implement the following flow table using (1) gates, (2) SR latch

x_1x_2

	00	01	11	10
a	a	c	b	a
b	a	c	b	b
c	a	c	c	b
d	a	d	d	a

Solution:

(1) using gates

Given, $x_1, x_2 \rightarrow$ inputs

$a, b, c, d \rightarrow$ states

$a = 00$ $c = 11$

$b = 01$ $d = 10$

$\rightarrow$ let $y_1, y_2 \rightarrow$ internal states

Thus the transition table is given as:

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	00	11	01	00
01	00	11	01	01
11	00	11	11	01
10	00	11	11	00

Transition table

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	1	0	0
01	0	1	0	0
11	0	1	1	0
10	0	1	1	0

map for y_1

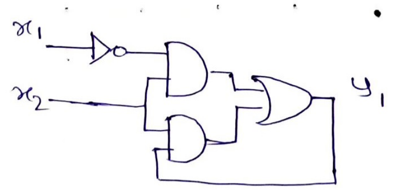
$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	1	1	0
01	0	1	1	1
11	0	1	1	1
10	0	1	1	0

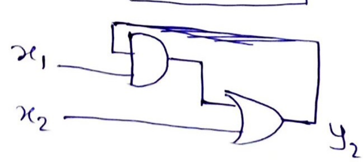
map for y_2

$\rightarrow$ Thus, $y_1 = \bar{x}_1 x_2 + y_1 x_2$

& $y_2 = x_2 + x_1 y_2$

$\rightarrow$ Thus implimentation is given as.





(2) using SR latch

- Excitation table SR latch is :

(Nor Gate S-R latch)

Q_n	Q_n^+	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

→ Using the excitation table the y_1 & y_2 maps can be converted to SR as follows :

S_1:

y_1y_2 \ x_1x_2	00	01	11	10
00	0	1	0	0
01	0	1	0	0
11	0	x	x	0
10	0	x	x	0

R_1:

y_1y_2 \ x_1x_2	00	01	11	10
00	x	0	x	x
01	x	0	x	x
11	1	0	0	1
10	1	0	0	1

y_1 ⇒ $S_1 = \overline{x_1} x_2$, $R_1 = \overline{x_2}$

S_2:

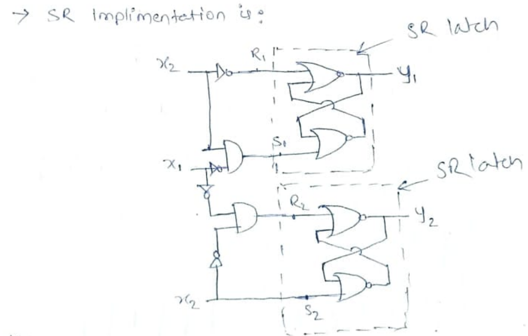
y_1y_2 \ x_1x_2	00	01	11	10
00	0	1	1	0
01	0	x	x	x
11	0	x	x	x
10	0	1	1	0

R_2:

y_1y_2 \ x_1x_2	00	01	11	10
00	x	0	0	x
01	1	0	0	0
11	1	0	0	0
10	x	0	0	x

y_2 ⇒ $S_2 = x_2$, $R_2 = \overline{x_1}\,\overline{x_2}$

→ SR Implimentation is :



16-3

9. Implement the following flow table using (1) gates, (2) D latch

	x_1x_2 = 00	01	11	10
a	a	a	b	d
b	b	c	a	d
c	b	c	d	d
d	a	d	c	d

Solution:

(1) using gates

Given, $x_1, x_2 \rightarrow$ inputs $\qquad a = 00 \quad c = 11$

$a, b, c, d \rightarrow$ states $\qquad b = 01 \quad d = 10$

$\rightarrow$ let y_1 & $y_2 \rightarrow$ Internal states

$\Rightarrow$ transition table :

y_1y_2 \ x_1x_2	00	01	11	10
00	00	00	01	10
01	01	11	00	10
11	01	11	10	10
10	00	10	11	10

Transition table

y_1y_2 \ x_1x_2	00	01	11	10
00	0	0	0	1
01	0	1	0	1
10	0	1	1	1
11	0	1	1	1

Y_1 map

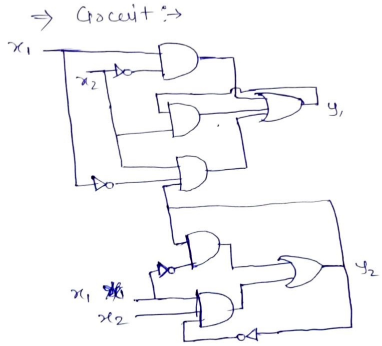
y_1y_2 \ x_1x_2	00	01	11	10
00	0	0	1	0
01	1	1	0	0
11	1	1	0	0
10	0	0	1	0

Y_2 map

$$\Rightarrow \quad Y_1 = x_1\overline{x_2} + x_2 y_1 + y_2 x_2 \overline{x_1}$$

$$Y_2 = \overline{x_1}\, y_2 + \overline{y_2}\, x_1 x_2$$

$\Rightarrow$ Circuit :$\rightarrow$



(2) using D latch

- Excitation table of D-latch is:

Q_n	Q_n^+	D
0	0	0
0	1	1
1	0	0
1	1	1

→ Using the excitation table the y_1 & y_2 maps can be converted to D_1 & D_2 as follows:

D1 map

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	0	0	1
01	0	1	0	1
11	0	1	1	1
10	0	1	1	1

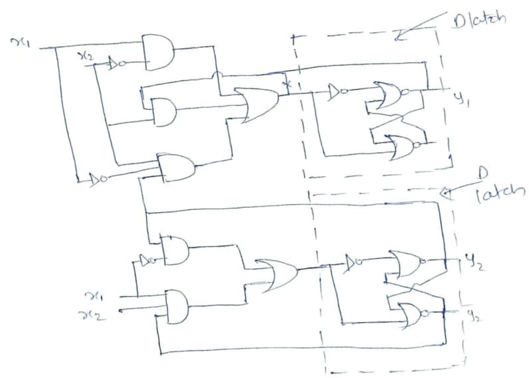
D2 map

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	0	1	0
01	1	1	0	0
11	1	1	0	0
10	0	0	1	0

$$\Rightarrow D_1 = x_1\bar{x}_2 + y_1x_2 + y_2\bar{x}_1x_2$$

$$D_2 = y_2\bar{x}_1 + \bar{y}_2x_1x_2$$

→ D latch implimentation is



10. Design an asynchronous sequential circuit with two inputs (X_1 and X_2) and one output (Z). Whenever the input X_2 is 1, the input X_1 is transferred to output Z. When X_2 is 0; the output Z does not change for a change in X_1. Consider the fundamental mode of operation.

Solution:

Given,

$x_1, x_2 \rightarrow$ inputs $\qquad$ $z \rightarrow$ output

$x_2 = 1 \Rightarrow Z = x_1$

$x_2 = 0 \Rightarrow Z = z$

States	Input		Output	Comments
	x_1	x_2	z	
a	0	0	0	let initial condⁿ 000
b	0	1	0	After state a
c	1	0	0	After state a
d	1	1	1	After states b, c
e	1	0	1	After state d
f	0	0	1	After state e

- Specification of total states

→ Thus the primitive flow table can be drawn as:

	x_1x_2 = 00	01	11	10
a	a, 0	b, –	–, –	c, –
b	a, –	b, 0	d, –	–, –
c	a, –	–, –	d, –	c, 0
d	–, –	b, –	d, 1	e, –
e	f, –	–, –	d, –	e, 1
f	f, 1	b, –	–, –	e, –

→ The obtained flow table can be reduced to two variable's by grouping similar states:

	00	01	11	10
$S_0 = abc$	S_0, 0	S_0, 0	S_1, —	S_0, 0
$S_1 = def$	S_1, 1	S_0, —	S_1, 1	S_1, 1

→ Now assigning binary values to state

• transition diagram:

→ No extra states are needed (race free)

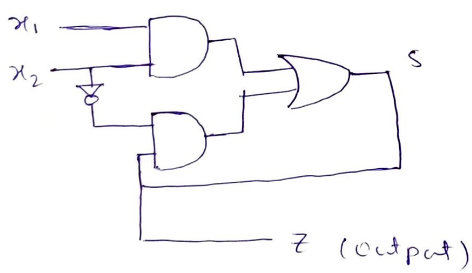
→ The transition table & output map be written as

$S \backslash x_1x_2$	00	01	11	10
0	0	0	1	0
1	1	0	1	1

Z $S \backslash x_1x_2$	00	01	11	10
0	0	0	*	0
1	1	*	1	1

$S^+ = x_1x_2 + S\bar{x}_2$; $Z = S$. (output)

→ Implimentation using gates



Practice problems

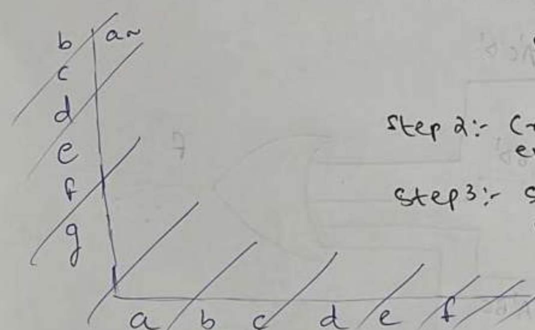
11. Use the implication-table method, to reduce the state table.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1

Solution:

(Q1)

Present state	Next state $x=0$	Next state $x=1$	Output $x=0$	Output $x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1



Step 1:- draw the implication table

Step 2:- Create one-one correspondence between the states.

Step 3:- Strike of the states whose outputs are not equal.

Two states can only equal if and only if, their outputs are equal w.r.t. input.

b	a~c b~d					
c	b~d	a~c				
d	a~e ~~b~f~~	c~e ~~f~d~~	a~e ~~f~d~~			
e	~~b~f~~	a~c ~~f~d~~	~~d~f~~	a~e		
f	a~g ~~b~f~~	c~g ~~d~f~~	a~g ~~d~f~~	e~g	a~g	
g	~~b~f~~	a~c ~~f~d~~	~~d~f~~	a~e	X	a~g
	a	b	c	d	e	f

We have removed all the b~f & ~~b~d~~ f~d correspondences.

Hence the remaining states, which are equal,

$a = c = g = e$

& $b = d$

12. Reduce the number of states in the state table. Use implication table.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	e	a	0	0
b	c	d	0	0
c	e	f	0	0
d	h	b	1	0
e	c	d	0	0
f	e	a	1	1
g	h	g	0	1
h	h	b	1	0

Solution:

(Q2)

Present state	Next state $x=0$	Next state $x=1$	Output $x=0$	Output $x=1$
a	e	a	0	0
b	c	d	0	0
c	e	f	0	0
d	h	b	1	0
e	c	d	0	0
f	e	a	1	1
g	h	g	0	1
h	h	b	1	0

b	c~e a~d						
c	a~f	c~e d~f					
d	e~h a~b	c~h d~b	e~h b~f				
e	c~e a~d	c~e d~d	c~e d~f	h~c b~d			
f	e~e a~a	c~e f~a	a~f	h~e b~a	c~e d~a		
g	e~h a~g	f~h g~d	e~h f~g	b~g	c~h d~g	e~h a~g	
h	e~h a~b	c~h b~d	e~h b~f	h~h b~b	f~h b~d	e~h a~b	g~b
	a	b	c	d	e	f	g

from the table
b = e
& d = h as both
output & next
states are equal.

$\boxed{\begin{matrix} b = e \\ d = h \end{matrix}}$

13. Merge each of the primitive flow tables shown below. Proceed as follows:

	00	01	11	10
a	a,0	b,–	–,–	e,–
b	a,–	b,0	c,–	–,–
c	–,–	d,–	c,0	h,–
d	a,–	d,1	–,–	–,–
e	a,–	–,–	f,–	e,0
f	–,–	g,–	f,0	h,–
g	a,–	g,0	–,–	–,–
h	a,–	–,–	–,–	h,0

(I)

	00	01	11	10
a	–,–	d,–	b,–	a,0
b	–,–	d,–	b,1	c,–
c	f,–	–,–	b,–	c,1
d	f,–	d,0	e,–	–,–
e	–,–	d,–	e,0	c,–
f	f,0	d,–	–,–	a,–

(II)

(a) Find all compatible pairs by means of an implication table.

(b) Find the maximal compatibles by means of a merger diagram.

(c) Find a minimal set of compatibles that covers all the states and is closed

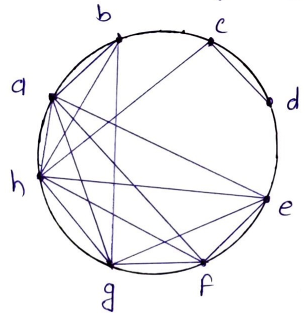
Solution:

(I) Implication table

	a	b	c	d	e	f	g
b	✓						
c	bd, eh ✗	bd ✗					
d	✗	✗	✓				
e	✓	cf ✗	cf, he ✗	✓			
f	bg, eh	bg, cf ✗	dg, cf ✗	dg ✗	eh		
g	bg	bg	dg ✗	✗	✓	✓	
h	eh	✓	✓	✓	eh	✓	✓

→ Hence, compatible pairs are:
ab, ae, af, ag, ah, bg, bh, cd, ch, ef, eg, eh, fg, fh, gh

- Merger diagram:



→ Maximal compatibles are: ab, cd, ch, efgh

- closure table:

Compatibles	ab	cd	ch	efgh
Implied states	-	-	-	eh

→ implied state 'eh' already included in maximal compatible. Hence, minimal set of compatibles are:

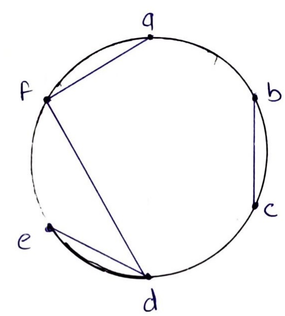
ab, cd, ch, efgh

(II) Implication table:-

	a	b	c	d	e
b	ae ×				
c	ac×	✓			
d	be×	be×	be×		
e	be, ac ×	×	be×	✓	
f	✓	ac×	ac×	✓	ac×

→ Thus the all compatible pairs are: df, af, de, cb

Merger diagram



→ Maximum compatibles are: bc, de, af

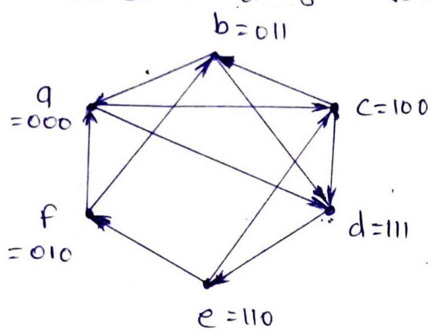
→ Here the minimal collection of compatibles is also the maximal compatible set.

14. Find a critical race-free state assignment for the reduced flow table shown below:

	00	01	11	10
a	a	d	a	c
b	a	b	b	d
c	d	c	b	c
d	d	d	e	d
e	f	c	e	c
f	f	b	a	f

Solution:

• Transition diagram for given flow table is:



• Transition diagram for the given flow table

→ Two extra states (g=001 & h=101) can be utilized for race free assignment

→ Hence the binary assignment table is given as

y_3 \ y_1y_2	00	01	11	10
0	a	g	b	f
1	c	h	d	e

15. Consider the reduced flow table shown in below Fig.

	00	01	11	10
a	a	c	a	d
b	a	b	c	b
c	c	c	c	d
d	d	b	a	d

(a) Obtain the transition diagram, and show that three state variables are needed for a race-free binary state assignment.

(b) Obtain the expanded flow table using the multiple-row method of assignment.

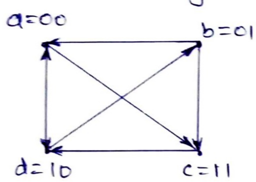
(c) Obtain the expanded flow table using the four-row method of assignment.

(d) Compare the number of states required in (b) and (c)

Solution:

(a)

- Transition diagram for given flow table is



→ The transitions from state a to c & d to b requires two bits change, which can arise race condition

→ To avoid it, extra states need to be added

→ Thus three-state variables are needed for the ~~to~~ race-free binary state assignment.

(b) • Multiple-row method:-

let, $a_1 = 000$ $\quad b_1 = 001$ $\quad c_1 = 011$ $\quad d_1 = 010$

$a_2 = 111$ $\quad b_2 = 110$ $\quad c_2 = 100$ $\quad d_2 = 101$

→ Binary state assignment table is given as.

$y_3 \backslash y_1 y_2$	00	10	11	01
0	a_1	b_1	c_1	d_1
1	c_2	d_2	a_2	b_2

→ Thus expanded flow table is given as.

	00	01	11	10
a_1	a_1	c_2	a_1	d_1
a_2	a_2	c_1	a_2	d_2
b_1	a_1	b_1	c_1	b_1
b_2	a_2	b_2	c_2	b_2
c_1	c_1	c_1	c_1	d_1
c_2	c_2	c_2	c_2	d_2
d_1	d_1	b_2	a_1	d_1
d_2	d_2	b_1	a_2	d_2

→ Total 8-states are required

(c) Four-row method:

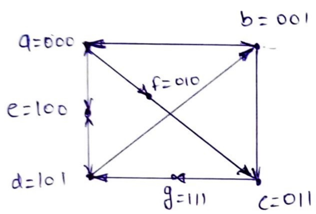
let a = 000 b = 001 c = 011 & d = 101

Three extra states e = 100, f = 010 & g = 111

→ Thus the binary state assignment table is given as,

$y_3 \backslash y_1 y_2$	00	01	11	10
0	a	b	c	f
1	e	d	g	

→ The modified transition diagram can be given as



→ Thus the expanded flow table is given as

	00	01	11	10
a	a	f	a	e
b	a	b	c	b
c	c	c	c	g
d	d	b	e	d
e	-	-	a	d
f	-	c	-	-
g	-	-	-	d

→ Total 7-states are required

(d) 7-states are required in Four-row method compared to 8-states in multiple row method.

16. Design an asynchronous sequential circuit using SR latch with two inputs (X_1 and X_2) and one output (Z), satisfying the following conditions:

(a) The inputs X_1 and X_2 never change simultaneously.

(b) The output z is 1 if $X_1 = 1$ and the X_2 changes from 0 to 1. Otherwise, the output z remains unchanged under any other input condition.

Solution:

Given,

$x_1, x_2 \longrightarrow$ inputs $\quad z \rightarrow$ output

$x_1 = 1$ & $x_2 = 0 \rightarrow 1 \Rightarrow z = 1$

• Specification of total states:

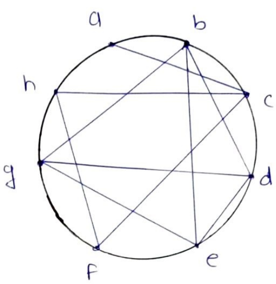
States	Inputs x_1	x_2	Output z	Comments
a	1	0	0	Initial output z=0
b	1	1	1	After a state
c	0	0	0	After state a
d	0	1	1	After state b
e	1	0	1	After state b
f	0	1	0	After state c
g	0	0	1	After state d
h	1	1	0	After state f

→ The primitive flow table obtained as:

	$x_1 x_2$ = 00	01	11	10
a	c, -	-, -	b, -	a, 0
b	-, -	d, -	b, 1	e, -
c	c, 0	f, -	-, -	a, -
d	g, -	d, 1	b, -	-, -
e	g, -	-, -	b, -	e, 1
f	c, -	f, 0	h, -	-, -
g	g, 1	d, -	-, -	e, -
h	-, -	f, -	h, 0	a, -

→ Implication table :-

b	ae ×						
c	✓	df ae ×					
d	cg ×	✓	cg, df ×				
e	×	✓	cg, ae ×	✓			
f	bh ×	df, bh ×	✓	×	cg, bh ×		
g	cg, ae ×	✓	×	✓	✓	cg, df ×	
h	bh ×	×	✓	df, bh ×	bh, ae ×	✓	ae, df ×
	a	b	c	d	e	f	g



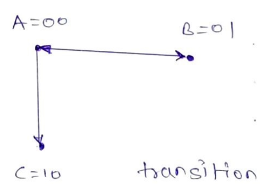
→ Compatible pairs are:
ac, db, be, bg, cf, ch, de, dg, eg, fh

→ maximal compatibles are: ac, cfh, bdeg

→ The minimal set of compatibles is also the maximal compatible set because covering & closed condition are satisfied.

→ Hence the reduced flow table can be given as:

	x_1x_2 00	01	11	10
A = ac	A, 0	B, –	C, –	A, 0
B = cfh	B, 0	B, 0	B, 0	A, –
C = bdeg	C, 1	C, 1	C, 1	C, 1



transition diagram

• State Assignment :-

→ Transition diagram can be given as,

• transition table :- let $y_1 y_2 \rightarrow$ internal state

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	00	01	10	00
01	01	01	01	00
11	xx	xx	xx	xx
10	10	10	10	10

Transition table

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	X	X	X	X
10	1	1	1	1

Y_1 map

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	1	0	0
01	1	1	1	0
11	X	X	X	X
10	0	0	0	0

Y_2 map

$y_1y_2 \backslash x_1x_2$	00	01	10	11
00	0	X	X	0
01	0	0	0	X
11	X	X	X	X
10	1	1	1	1

output Z map

$\Rightarrow$

$$Y_1 = y_1 + \bar{y}_2 x_1 x_2$$

$$Y_2 = \bar{y}_1 \bar{x}_1 x_2 + y_2 \bar{x}_1 + y_2 x_2$$

$$Z = y_1$$

Q_n	Q_n^+	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

$\longrightarrow$ Implimentation using SR latch:

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	x	x	x	x
10	x	x	x	x

S_1

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	x	x	0	x
01	x	x	x	x
11	x	x	x	x
10	0	0	0	0

R_1

$\Rightarrow$

$$S_1 = \bar{y}_2 x_1 x_2$$

$$R_1 = 0$$

$$S_2 = \bar{y}_1 \bar{x}_1 x_2$$

$$R_2 = x_1 \bar{x}_2$$

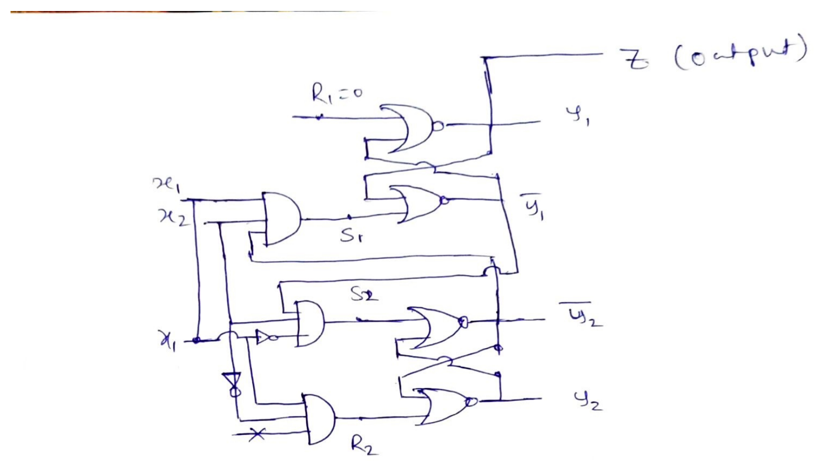
$$Z = y_1$$

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	0	1	0	0
01	x	x	x	0
11	x	x	x	x
10	0	0	0	0

S_2

$y_1y_2 \backslash x_1x_2$	00	01	11	10
00	x	0	x	x
01	0	0	0	1
11	x	x	x	x
10	x	x	x	x

R_2



17. A traffic light is installed at the junction of a railroad and a road. The light is controlled by two switches in the rails placed 1 mile apart on either side of the junction. A switch is turned on when the train is over it and is turned off otherwise. The traffic light changes from green (logic 0) to red (logic 1) when the beginning of the train is 1 mile from the junction. The light changes back to green when the end of the train is 1 mile away from the junction. Assume that the length of the train is less than 2 miles. Consider the fundamental mode of operation.

    (a) Obtain a primitive flow table for the circuit.

    (b) Show that the flow table can be reduced to four rows.

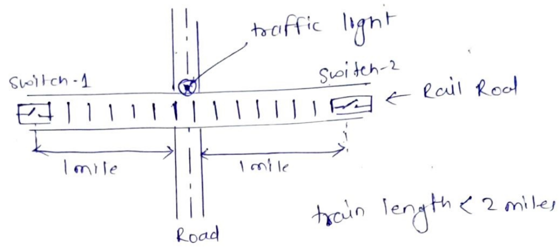
    (c) Complete the design of the circuit using D latch

Solution:

Q)

Soln:-

Given,



let $x_1 \rightarrow$ switch-1, $x_2 \rightarrow$ switch-2 } inputs $z \rightarrow$ light } output

① before & end of passing of train $\Rightarrow x_1x_2 = 00$ & $z = 0$

② train over the switches $\Rightarrow x_1 x_2 = 10$ or $= 01$ $\Rightarrow z = 1$

③ train between the switches $\Rightarrow x_1x_2 = 00 \Rightarrow z = 1$

④ train may come from right or left side.

⑤ $x_1x_2 = 11$ Not possible $\Leftarrow$ train length < 2 miles

→ • Specifications of the states:

States	Input x_1	Input x_2	Output z	Comments
a	0	0	0	No train, End of train
b	1	0	1	train over sw-1, After a
c	0	1	1	train over sw-2, After a
d	0	0	1	train betn sw's, After b, c
e	0	1	1	train over sw-2, After d
f	1	0	1	train over sw-1, After d

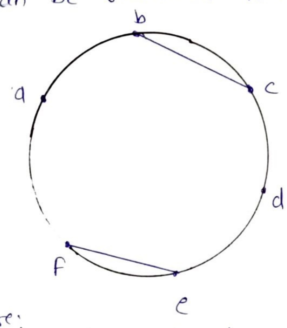
→ Primitive flow table:

	00	01	11	10
a	a,0	c,-	-,-	b,-
b	d,-	-,-	-,-	b,1
c	d,-	c,1	-,-	-,-
d	d,1	e,-	-,-	f,-
e	a,-	e,1	-,-	-,-
f	a,-	-,-	-,-	f,1

(x_1x_2)

→ This flow table can be reduced further:

	a	b	c	d	e
b	ad ×				
c	ad ×	✓			
d	×	bf ×	ce ×		
e	ce ×	ad ×	ad ce ×	ad ×	
f	bf ×	ad bf ×	ad ×	ad ×	✓



→ compatible pairs are:
a, bc, d, ef

→ maximal compatibles are: a, bc, d, ef

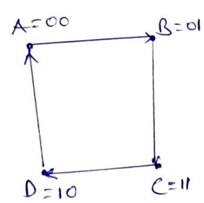
→ minimal set of compatibles = maximal compatibles
⇐ both covering & closed cond^n s satisfied

→ The reduced flow ~~char~~ table is:-

	00	01	11	10
A = a	A,0	B,-	-,-	B,-
B = bc	C,-	B,1	-,-	B,1
C = d	C,1	D,-	-,-	D,-
D = ef	A,-	D,1	-,-	D,1

(x_1x_2)

⇒ transition diagram ⇒



→ Transition table:-

y_1y_2 \ x_1x_2	00	01	11	10
00	00	01	xx	01
01	11	01	xx	01
11	11	10	xx	10
10	00	10	xx	10

transition table

y_1y_2 \ x_1x_2	00	01	11	10
00	0	0	x	0
01	1	0	x	0
11	1	1	x	1
10	0	1	x	1

y_1 map

y_1y_2 \ x_1x_2	00	01	11	10
00	0	1	x	1
01	1	1	x	1
11	1	0	x	0
10	0	0	x	0

y_2 map

y_1y_2 \ x_1x_2	00	01	11	10
00	0	x	x	x
01	x	1	x	1
11	1	x	x	x
10	x	1	x	1

output map

y_1y_2 \ x_1x_2	00	01	11	10
00	0	0	x	0
01	1	0	x	0
11	1	1	x	1
10	0	1	x	1

D_1 map

y_1y_2 \ x_1x_2	00	01	11	10
00	0	1	x	1
01	1	1	x	1
11	1	0	x	0
10	0	0	x	0

D_2 map

→ Equations:

$$D_1 = y_2\overline{x_1}\,\overline{x_2} + y_1x_2 + y_1x_1$$

$$D_2 = y_2\overline{x_1}\,\overline{x_2} + \overline{y_1}x_2 + \overline{y_1}x_1$$

$$Z = y_1 + y_2$$

-> Implimentation: